Finite Element Modeling of Heat Conduction in a Microtube Contact Geometry:

2D/3D Steady and Transient Solutions with Convection, Prescribed Heat Flux, and a Thermal Paste Layer

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Abstract

Accurate temperature measurement on electrically heated microtubes is experimentally challenging because thermocouples sense a voltage difference while the tube itself may carry an electrical potential. A method associated with Hojin Ahn and collaborators improves attachment by forming a molded solder contact to maximize contact area and repeatability, with a thin thermal paste layer to control the interface. Because the geometry is extremely small, direct experimental observation of internal temperature distributions is limited, motivating a numerical study. This project develops finite element method (FEM) formulations and implementations for heat conduction in a small rectangular body containing a tube-contact cut. The following cases are solved: (i) 2D steady conduction (single material), (ii) 2D transient conduction (single material), (iii) 2D steady and transient conduction with an added thermal paste layer (two materials), and (iv) 3D steady and transient conduction with thermal paste. The governing model is the heat equation with piecewise material properties and mixed boundary conditions: prescribed heat flux (Neumann) at the tube-contact boundary, convection (Robin) at exposed surfaces, and symmetry (homogeneous Neumann) at the symmetry plane. Transient problems are integrated using Backward Euler, and 3D fields are inspected using internal slicing.

Keywords: Finite Element Method; Heat Conduction; Neumann Boundary; Robin boundary; Transient; Backward Euler; Thermal paste; Microtube

1 Introduction

Thermal measurements on microtubes often rely on thermocouples, which infer temperature through an electrical potential difference. When the tube is electrically heated (or otherwise carries a potential), attaching a thermocouple via solder can bias the measured voltage. A practical strategy reported in the literature uses a molded solder contact that increases contact surface area and improves repeatability, while a thin thermal paste layer controls the interface condition [1]. However, because the geometry is very small, it is difficult to experimentally observe the internal temperature distribution and to separate conduction paths from surface losses. Therefore, a numerical approach becomes useful for interpreting experiments and verifying the consistency of boundary conditions.

In this Math 543 (Boundary Value Problems) course project, the numerical workflow is:

1. 2D meshing on a half-domain (symmetry reduction);

2. 2D steady-state conduction (single material);
3. 2D transient conduction (single material);
4. 2D steady and transient conduction with a thermal paste layer;
5. 3D meshing by extrusion of the 2D cross-section and tetrahedralization;
6. 3D steady and transient conduction with thermal paste and internal slicing visualization.

2 Geometry and Model Parameters

2.1 Geometry

The base geometry is a rectangular body in cross-section, with a circular cut representing the tube-contact region. A thermal paste layer is modeled as a thin annulus surrounding the tube-contact arc/cylinder.

- Cross-section width: $W = 1.4\,\text{mm}$.
- Cross-section height: $H = 1.0\,\text{mm}$.
- 3D extrusion length: $L_z = 1.0\,\text{mm}$.
- Cut diameter: $d = 1.0\,\text{mm} \Rightarrow R_{\text{in}} = d/2 = 0.5\,\text{mm}$.
- Paste thickness: $t_p = 0.1\,\text{mm} \Rightarrow R_{\text{out}} = R_{\text{in}} + t_p = 0.6\,\text{mm}$.

A symmetry plane at $x = W/2$ is used to solve a half-domain and mirror results for full-geometry visualization.

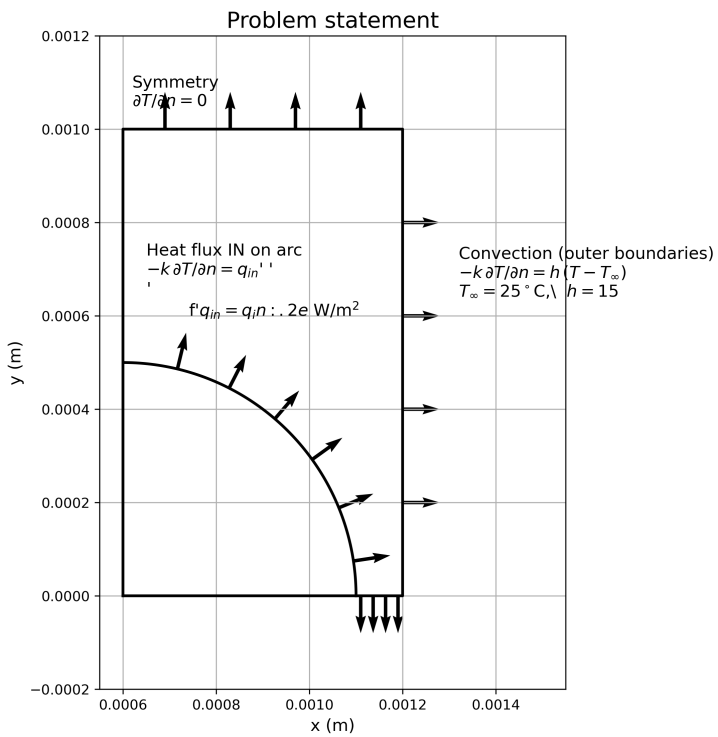

Figure 1: 2D problem definiton of the problem.

2.2 Material Properties

Two materials are used when paste is included:

- Solid: $k_s = 16\,\mathrm{W/(m \cdot K)}$, $\rho_s = 8000\,\mathrm{kg/m^3}$, $c_{p,s} = 500\,\mathrm{J/(kg \cdot K)}$.
- Thermal paste: $k_p = 6\,\mathrm{W/(m \cdot K)}$, $\rho_p = 2500\,\mathrm{kg/m^3}$, $c_{p,p} = 1000\,\mathrm{J/(kg \cdot K)}$.

2.3 Boundary and Time Parameters

- Prescribed heat flux on tube-contact boundary: $q_{\mathrm{in}} = 1.0 \times 10^3\,\mathrm{W/m^2}$.
- Convection on exposed surfaces: $h = 15\,\mathrm{W/(m^2 \cdot K)}$, $T_\infty = 25°\mathrm{C}$.
- Transient step size: $\Delta t = 0.5\,\mathrm{s}$.
- Final time: $t_{\mathrm{end}} = 1000\,\mathrm{s}$.

3 Governing Equations

This section introduces the mathematical model used across all cases in this project. Because the geometry is small and the dominant transport mechanism inside the solid and paste is conduction, the temperature field is modeled using the classical heat equation. The same governing equation is used in 2D and 3D; only the computational domain Ω and the boundary partition differ.

3.1 Energy Balance and Fourier's Law

The resulting heat conduction equation represents the standard diffusion model for solids and is widely used for micro-scale and macro-scale heat transfer analysis when convection and radiation are treated through boundary conditions [2, 3]. Let $T(\mathbf{x}, t)$ denote the temperature at position $\mathbf{x}$ and time t. For a stationary material with density ρ and specific heat c_p, conservation of energy written per unit volume states that the rate of change of internal energy equals the negative divergence of the heat flux:

$$\rho c_p \frac{\partial T}{\partial t} = -\nabla \cdot \mathbf{q}. \tag{1}$$

In this project, volumetric heat generation is not included because the electrical heating occurs in the microtube and enters the modeled domain through a boundary heat flux (Section 4).

The conductive heat flux vector $\mathbf{q}$ is related to the temperature gradient by Fourier's law,

$$\mathbf{q} = -k \nabla T, \tag{2}$$

where k is the thermal conductivity. Substituting (2) into (1) yields the transient heat conduction equation:

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) \quad \text{in } \Omega. \tag{3}$$

Equation (3) describes how temperature evolves in time as conduction redistributes energy within the domain. For constant k, it reduces to $\rho c_p\, \partial T/\partial t = k \nabla^2 T$; however, in this project k may vary spatially due to the thermal paste layer.

3.2 Steady-State Limit

For steady-state problems, the temperature does not change with time, i.e., $\partial T/\partial t = 0$. Taking this limit in (3) gives

$$\nabla \cdot (k \nabla T) = 0 \quad \text{in } \Omega, \tag{4}$$

which is the elliptic (Laplace-type) conduction equation used in the 2D steady and 3D steady simulations. In practice, steady and transient solutions are closely connected: transient simulations with implicit time stepping (Section **??**) approach the steady solution as $t \to \infty$ under time-independent boundary conditions.

3.3 Piecewise Material Model for the Thermal Paste Layer

When the thermal paste layer is included, the computational domain is split into the solid region Ω_s and paste region Ω_p:

$$\Omega = \Omega_s \cup \Omega_p, \qquad \Omega_s \cap \Omega_p = \emptyset. \tag{5}$$

Material coefficients are treated as piecewise constants,

$$k(\mathbf{x}) = \begin{cases} k_s, & \mathbf{x} \in \Omega_s, \\ k_p, & \mathbf{x} \in \Omega_p, \end{cases} \qquad \rho c_p(\mathbf{x}) = \begin{cases} \rho_s c_{p,s}, & \mathbf{x} \in \Omega_s, \\ \rho_p c_{p,p}, & \mathbf{x} \in \Omega_p. \end{cases} \tag{6}$$

This representation captures the dominant effect of the paste: it changes the effective conduction path between the microtube-contact boundary and the bulk solid. At the internal interface Γ_{sp} between the two regions, the physical continuity conditions are

$$T_s = T_p \quad \text{and} \quad -k_s \nabla T_s \cdot \mathbf{n} = -k_p \nabla T_p \cdot \mathbf{n}, \tag{7}$$

meaning temperature is continuous and heat flux is conserved across the interface. In the FEM implementation, these interface conditions are enforced naturally by assembling element contributions into a single global system with shared nodes along Γ_{sp}.

3.4 Model Scope and Assumptions

The model intentionally isolates conduction inside the body and paste while treating heat exchange with the environment through boundary conditions. Radiation is neglected because the characteristic temperatures and length scale in the present setup make convection the primary external mechanism in the intended operating range. Similarly, the internal fluid flow inside the microtube is not modeled here; instead, net effect of the heated wall of the microtube has represented by prescribing a heat flux into the contact boundary. With these assumptions, the governing equation (3) and its steady limit (4), together with appropriate boundary conditions, define a closed boundary value problem.

4 Boundary Conditions

The governing equations introduced in Section 3 require appropriate boundary conditions to represent the physical interaction between the solid, the microtube, and the surrounding environment. In the present problem, heat enters the domain through the microtube-contact surface, is redistributed by conduction within the solid and thermal paste, and is removed through convection on the exposed outer surfaces. A symmetry condition is additionally imposed to reduce the computational domain.

Let $\Gamma = \partial\Omega$ denote the boundary of the domain and $\mathbf{n}$ the outward unit normal. The outward normal conductive heat flux is defined as

$$q_n \equiv -k \nabla T \cdot \mathbf{n}. \tag{8}$$

The boundary Γ is partitioned into three disjoint parts:

$$\Gamma = \Gamma_{\text{in}} \cup \Gamma_{\text{conv}} \cup \Gamma_{\text{sym}},$$

corresponding to the microtube-contact surface, the exposed outer surfaces, and the symmetry plane, respectively.

4.1 Prescribed Heat Flux on the Microtube-Contact Boundary

On the microtube-contact boundary Γ_{in} (the inner arc in 2D and the inner cylindrical surface in 3D), the thermal interaction between the tube and the surrounding solid is modeled as a prescribed heat flux. This approach is appropriate because the electrical heating and internal fluid flow occur inside the tube, and their net thermal effect on the solid can be represented as a known surface heat input.

The Neumann boundary condition is therefore written as

$$-k\nabla T \cdot \mathbf{n} = q_{\text{in}} \quad \text{on } \Gamma_{\text{in}}, \tag{9}$$

where $q_{\text{in}} = 1.0e3W/m^2$ is the imposed heat flux. A positive value of q_{in} corresponds to heat flowing *into* the computational domain. This condition is identical in both 2D and 3D formulations and directly enters the weak form as a boundary integral.

4.2 Convective Heat Loss on Exposed Outer Surfaces

All outer surfaces that are exposed to the surrounding air are assumed to lose heat by convection. Radiative effects are neglected because of the small characteristic length scale and moderate temperature levels. The convective heat flux is modeled using Newton's law of cooling,

$$q_{\text{conv}} = h\left(T - T_\infty\right), \tag{10}$$

where $h = 15W/(m^2 \cdot K)$ is the convection coefficient and $T_\infty = 25°C$ is the ambient temperature.

Equating the conductive flux (8) to the convective flux (10) yields a Robin (mixed) boundary condition:

$$-k\nabla T \cdot \mathbf{n} = h\left(T - T_\infty\right) \quad \text{on } \Gamma_{\text{conv}}. \tag{11}$$

This boundary condition introduces both stiffness-like and load-like contributions in the finite element formulation. Physically, it represents the continuous removal of heat from the solid to the environment and prevents the temperature from increasing indefinitely under a constant heat flux input.

4.3 Symmetry Boundary Condition

The geometry and loading are symmetric with respect to a vertical plane passing through the centerline of the microtube. To exploit this symmetry and reduce computational cost, only half of the domain is solved numerically. The symmetry plane is denoted by Γ_{sym}.

Because no heat crosses the symmetry plane, the normal temperature gradient vanishes:

$$\nabla T \cdot \mathbf{n} = 0 \quad \text{on } \Gamma_{\text{sym}}. \tag{12}$$

This is a homogeneous Neumann boundary condition and is *naturally* satisfied in the weak formulation, meaning it does not generate additional boundary integrals. After solving the half-domain problem, the temperature field is mirrored to obtain the full geometry for visualization.

4.4 Summary of Boundary Conditions

For clarity, the complete boundary value problem is summarized as follows:

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) \qquad \text{in } \Omega, \tag{13}$$

$$-k \nabla T \cdot \mathbf{n} = q_{\text{in}} \qquad \text{on } \Gamma_{\text{in}}, \tag{14}$$

$$-k \nabla T \cdot \mathbf{n} = h\,(T - T_\infty) \qquad \text{on } \Gamma_{\text{conv}}, \tag{15}$$

$$\nabla T \cdot \mathbf{n} = 0 \qquad \text{on } \Gamma_{\text{sym}}. \tag{16}$$

These boundary conditions are used consistently in all 2D and 3D cases, with transient and steady behavior distinguished only by the presence or absence of the time derivative term.

5 Weak Formulation

The strong (differential) form of the heat conduction equation derived in Section 3, together with the boundary conditions described in Section 4, cannot be solved analytically for the present geometry and mixed boundary conditions. Therefore, a weak (variational) formulation is introduced as the foundation for the finite element method.

The weak form reduces differentiability requirements on the temperature field, naturally incorporates Neumann and Robin boundary conditions, and allows the problem to be solved on unstructured meshes.

5.1 Derivation of the Weak Form

Consider the transient heat equation

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) \quad \text{in } \Omega. \tag{17}$$

Let $w(\mathbf{x})$ be an admissible test (weighting) function that is sufficiently smooth. Multiplying (17) by w and integrating over the domain Ω yields

$$\int_\Omega w\, \rho c_p \frac{\partial T}{\partial t}\, d\Omega - \int_\Omega w\, \nabla \cdot (k \nabla T)\, d\Omega = 0. \tag{18}$$

The second integral contains second-order spatial derivatives of T. To lower the order of differentiation, integration by parts is applied using the divergence theorem:

$$\int_\Omega w\, \nabla \cdot (k \nabla T)\, d\Omega = -\int_\Omega k \nabla w \cdot \nabla T\, d\Omega + \int_\Gamma w\, k \nabla T \cdot \mathbf{n}\, d\Gamma. \tag{19}$$

Substituting (19) into (18) and using the definition $q_n = -k \nabla T \cdot \mathbf{n}$ gives

$$\int_\Omega w\, \rho c_p \frac{\partial T}{\partial t}\, d\Omega + \int_\Omega k \nabla w \cdot \nabla T\, d\Omega = \int_\Gamma w\, q_n\, d\Gamma. \tag{20}$$

5.2 Incorporation of Boundary Conditions

The boundary Γ is decomposed as

$$\Gamma = \Gamma_{\text{in}} \cup \Gamma_{\text{conv}} \cup \Gamma_{\text{sym}}.$$

On the symmetry boundary Γ_{sym}, the condition $\nabla T \cdot \mathbf{n} = 0$ implies $q_n = 0$. Therefore, this part of the boundary does not contribute to the right-hand side of the weak form.

On the microtube-contact boundary Γ_{in}, the prescribed heat flux condition $-k\nabla T \cdot \mathbf{n} = q_{\text{in}}$ leads to

$$\int_{\Gamma_{\text{in}}} w\, q_{\text{in}}\, d\Gamma. \tag{21}$$

On the exposed outer boundary Γ_{conv}, the Robin boundary condition

$$-k\nabla T \cdot \mathbf{n} = h\,(T - T_\infty)$$

yields two contributions:

- a term proportional to the unknown temperature T,
- a term proportional to the ambient temperature T_∞.

Substituting the boundary conditions into (20) results in the final weak form of the transient problem:

$$\int_\Omega w\, \rho c_p \frac{\partial T}{\partial t}\, d\Omega + \int_\Omega k\nabla w \cdot \nabla T\, d\Omega + \int_{\Gamma_{\text{conv}}} w\, h\, T\, d\Gamma = \int_{\Gamma_{\text{in}}} w\, q_{\text{in}}\, d\Gamma + \int_{\Gamma_{\text{conv}}} w\, h\, T_\infty\, d\Gamma. \tag{22}$$

5.3 Steady-State Weak Form

For steady-state heat conduction, the temperature no longer depends on time, and the transient term vanishes. The weak form reduces to

$$\int_\Omega k\nabla w \cdot \nabla T\, d\Omega + \int_{\Gamma_{\text{conv}}} w\, h\, T\, d\Gamma = \int_{\Gamma_{\text{in}}} w\, q_{\text{in}}\, d\Gamma + \int_{\Gamma_{\text{conv}}} w\, h\, T_\infty\, d\Gamma. \tag{23}$$

This formulation is valid for both single-material and multi-material domains. When a thermal paste layer is present, the conductivity k and volumetric heat capacity ρc_p are treated as piecewise constants and are evaluated element-wise during numerical integration.

5.4 Remarks on the Weak Form

The weighted residual and Galerkin procedures employed here follow classical finite element formulations for diffusion-type problems. Such approaches are well established for heat transfer applications and provide a consistent framework for incorporating Neumann and Robin boundary conditions [4, 5]. Several important features of the weak formulation are worth emphasizing:

- Neumann and symmetry boundary conditions appear naturally and require no additional constraints.
- Robin boundary conditions contribute both stiffness-like and load-like terms.
- The formulation is dimension-independent and applies directly to both 2D and 3D geometries.
- The weak form provides a natural transition to finite element discretization using standard shape functions.

6 Finite Element Discretization

The weak formulations derived in Section 5 are discretized using the finite element method (FEM). The spatial domain is subdivided into finite elements, and the temperature field is approximated by a finite-dimensional space spanned by nodal basis (shape) functions. This procedure converts the continuous boundary value problem into a system of algebraic equations that can be solved numerically.

6.1 Spatial Approximation

Let $\{N_j(\mathbf{x})\}_{j=1}^{N}$ denote the standard finite element shape functions associated with the N nodes of the mesh. The temperature field is approximated as

$$T(\mathbf{x}, t) \approx \sum_{j=1}^{N} N_j(\mathbf{x})\, T_j(t), \tag{24}$$

where $T_j(t)$ are the unknown nodal temperatures.

In the Galerkin method, the test function w is chosen from the same space:

$$w(\mathbf{x}) = N_i(\mathbf{x}). \tag{25}$$

This choice ensures consistency and symmetry of the resulting system matrices.

6.2 Discrete Weak Form

Substituting the finite element approximation (24) into the transient weak form (22) and choosing $w = N_i$ yields

$$\sum_{j=1}^{N} \left[\int_{\Omega} \rho c_p\, N_i N_j\, d\Omega\, \frac{dT_j}{dt} + \int_{\Omega} k\, \nabla N_i \cdot \nabla N_j\, d\Omega\, T_j + \int_{\Gamma_{\text{conv}}} h\, N_i N_j\, d\Gamma\, T_j \right] = f_i, \tag{26}$$

where the right-hand-side load term is

$$f_i = \int_{\Gamma_{\text{in}}} N_i\, q_{\text{in}}\, d\Gamma + \int_{\Gamma_{\text{conv}}} h\, T_\infty\, N_i\, d\Gamma. \tag{27}$$

6.3 Matrix Form

Equation (26) can be written compactly in matrix form as

$$\mathbf{M}\,\dot{\mathbf{T}} + (\mathbf{K} + \mathbf{K}_h)\,\mathbf{T} = \mathbf{f}_q + \mathbf{f}_h, \tag{28}$$

where:

$$M_{ij} = \int_\Omega \rho c_p \, N_i N_j \, d\Omega \qquad \text{(mass matrix)}, \tag{29}$$

$$K_{ij} = \int_\Omega k \, \nabla N_i \cdot \nabla N_j \, d\Omega \qquad \text{(conduction stiffness matrix)}, \tag{30}$$

$$(K_h)_{ij} = \int_{\Gamma_{\text{conv}}} h \, N_i N_j \, d\Gamma \qquad \text{(convection boundary matrix)}, \tag{31}$$

$$(f_q)_i = \int_{\Gamma_{\text{in}}} N_i \, q_{\text{in}} \, d\Gamma \qquad \text{(heat flux load vector)}, \tag{32}$$

$$(f_h)_i = \int_{\Gamma_{\text{conv}}} h \, T_\infty \, N_i \, d\Gamma \qquad \text{(convection load vector)}. \tag{33}$$

6.4 Steady-State System

For steady-state heat conduction, the time derivative vanishes, $\dot{\mathbf{T}} = \mathbf{0}$, and (28) reduces to

$$(\mathbf{K} + \mathbf{K}_h) \, \mathbf{T} = \mathbf{f}_q + \mathbf{f}_h. \tag{34}$$

This linear system is solved directly for the nodal temperature vector $\mathbf{T}$.

6.5 Multi-Material Treatment

When the thermal paste layer is included, the conductivity k and volumetric heat capacity ρc_p vary spatially. In the finite element framework, this is handled naturally by evaluating material properties at the element level:

- elements belonging to the solid use $(k_s, \rho_s c_{p,s})$,
- elements belonging to the thermal paste use $(k_p, \rho_p c_{p,p})$.

The global matrices $\mathbf{M}$ and $\mathbf{K}$ are assembled by summing contributions from all elements, ensuring continuity of temperature and heat flux across material interfaces without imposing additional interface conditions.

6.6 Remarks on the Discrete System

The finite element discretization leads to a system with the following properties:

- The mass matrix $\mathbf{M}$ is symmetric and positive definite.
- The stiffness matrix $\mathbf{K}$ is symmetric and positive semi-definite.
- The convection matrix $\mathbf{K}_h$ adds boundary-induced coupling and improves numerical stability.
- The formulation applies identically in two and three dimensions.

The resulting system provides the starting point for temporal discretization, which is addressed in the next section using an implicit Backward Euler scheme.

7 Time Integration Scheme

The transient heat conduction problems considered in this study are governed by the semi-discrete system obtained from the finite element discretization:

$$\mathbf{M}\,\dot{\mathbf{T}} + (\mathbf{K} + \mathbf{K}_h)\,\mathbf{T} = \mathbf{f}_q + \mathbf{f}_h, \tag{35}$$

where $\mathbf{T}(t)$ is the vector of nodal temperatures, $\mathbf{M}$ is the mass matrix, $\mathbf{K}$ is the conduction stiffness matrix, and $\mathbf{K}_h$ accounts for convection boundary effects.

To advance the solution in time, an implicit Backward Euler scheme is employed.

7.1 Motivation for Implicit Time Integration

Heat conduction problems are typically diffusion-dominated and may involve fine meshes, high thermal conductivities, or thin layers such as the thermal paste considered here. Under such conditions, explicit time integration schemes impose severe restrictions on the time step size for stability. Implicit time integration schemes such as Backward Euler are commonly used for transient heat conduction problems because of their unconditional stability and robustness for diffusion-dominated systems [3].

The Backward Euler method is chosen because:

- it is unconditionally stable for linear diffusion problems,
- it allows relatively large time steps without numerical instability,
- it naturally damps high-frequency numerical oscillations,
- it is well-suited for reaching steady-state solutions through transient marching.

These properties are especially important for three-dimensional simulations with multi-material interfaces.

7.2 Backward Euler Discretization

Let t^n and $t^{n+1} = t^n + \Delta t$ denote consecutive time levels. The time derivative of the temperature vector is approximated by

$$\dot{\mathbf{T}}(t^{n+1}) \approx \frac{\mathbf{T}^{n+1} - \mathbf{T}^n}{\Delta t}. \tag{36}$$

Substituting (36) into (35) and evaluating all terms at time t^{n+1} yields

$$\mathbf{M}\frac{\mathbf{T}^{n+1} - \mathbf{T}^n}{\Delta t} + (\mathbf{K} + \mathbf{K}_h)\,\mathbf{T}^{n+1} = \mathbf{f}_q + \mathbf{f}_h. \tag{37}$$

Rearranging terms gives the linear system solved at each time step:

$$\left(\frac{\mathbf{M}}{\Delta t} + \mathbf{K} + \mathbf{K}_h\right)\mathbf{T}^{n+1} = (\mathbf{f}_q + \mathbf{f}_h) + \frac{\mathbf{M}}{\Delta t}\mathbf{T}^n. \tag{38}$$

7.3 Implementation Details

In the present work, the time step size is chosen as

$$\Delta t = 0.5s, \tag{39}$$

and the solution is advanced up to a final simulation time of

$$t_{\text{end}} = 1000s. \tag{40}$$

Because all material properties and boundary conditions are time-independent, the system matrix

$$\mathbf{A} = \frac{\mathbf{M}}{\Delta t} + \mathbf{K} + \mathbf{K}_h \tag{41}$$

remains constant throughout the simulation. This allows the matrix to be assembled and factorized once, after which only the right-hand side vector changes at each time step.

7.4 Initial Condition

The transient simulations are initialized from a uniform temperature field equal to the ambient temperature:

$$T(\mathbf{x}, 0) = T_\infty = 25°C. \tag{42}$$

This choice corresponds to a physically realistic scenario in which the body is initially in thermal equilibrium with the surroundings before the heat flux is applied.

7.5 Relation to Steady-State Solution

As $t \to \infty$, the transient solution obtained from (38) approaches the steady-state solution of (34). In practice, the steady temperature field is reached once

$$\|\mathbf{T}^{n+1} - \mathbf{T}^n\| \ll \|\mathbf{T}^n\|, \tag{43}$$

or when the maximum temperature change per time step becomes negligible.

This property allows the transient solver to be used not only for time-dependent analysis, but also as a robust numerical pathway to the steady-state solution.

8 Numerical Implementation and Meshing Strategy

The numerical solution of the heat conduction problem is carried out using a custom finite element implementation based on unstructured meshes.Due to the curved geometry, mixed boundary conditions, and presence of thin multi-material regions, closed-form analytical solutions are not available for the present problem. Finite element methods therefore provide a practical and flexible approach for resolving the resulting temperature fields [4]. Both two-dimensional and three-dimensional cases are considered, with increasing geometric and physical complexity. Special care is taken to ensure adequate spatial resolution near the tube-contact region and across the thin thermal paste layer, where large temperature gradients are expected.

8.1 Dimensional Consistency and Scaling

Although the geometric dimensions are specified in millimeters for clarity, all computations are performed in SI units. Accordingly, all geometric coordinates are converted to meters before assembly, ensuring consistency with the thermal parameters k, ρ, c_p, q_{in}, and h. This avoids numerical inconsistencies while preserving intuitive interpretation of the geometry.

8.2 Two-Dimensional Mesh Generation

The two-dimensional computational domain consists of a rectangular cross-section with a circular cut representing the microtube contact. To reduce computational cost while preserving physical symmetry, only half of the domain is discretized, with a symmetry boundary imposed along the vertical centerline.

An unstructured triangular mesh is generated using local refinement controls. Element sizes are reduced near the tube-contact arc to accurately resolve the prescribed heat flux boundary and the resulting steep temperature gradients. A coarser mesh is used away from the contact region, where the temperature field varies more smoothly.

Figure **??** shows the resulting half-domain mesh, illustrating the local refinement near the inner arc of radius $R_{\text{in}} = 0.5$ mm.

8.3 Mesh Extension for the Thermal Paste Layer

When the thermal paste layer is included, the geometry is augmented by an additional annular region surrounding the tube-contact arc. This region extends from $R_{\text{in}} = 0.5$ mm to $R_{\text{out}} = 0.6$ mm and is assigned distinct thermal properties.

The mesh is constructed such that element boundaries conform to the material interface, ensuring that the piecewise conductivity and heat capacity are captured correctly during assembly. No special interface conditions are required, as continuity of temperature and heat flux is naturally enforced by the finite element formulation.

The resulting two-material mesh is shown in Figure **??**.

8.4 Three-Dimensional Mesh Construction

The three-dimensional geometry is obtained by extruding the two-dimensional cross-section along the out-of-plane direction over a length $L_z = 1.0$ mm. The extrusion is performed using a fixed number of layers to ensure a uniform resolution in the z-direction.

Following extrusion, the domain is discretized using unstructured tetrahedral elements. The mesh density in the x–y plane is inherited from the two-dimensional mesh, while the extrusion ensures consistent element aspect ratios. The half-domain is mirrored across the symmetry plane to obtain the full geometry for visualization and post-processing.

A surface view of the full three-dimensional mesh is shown in Figure **??**.

8.5 Boundary Classification and Assembly

After mesh generation, boundary facets are automatically classified into three groups:

- tube-contact boundaries, where a prescribed heat flux $q_{\text{in}} = 1.0 \times 10^3$ W/m^2 is applied;
- exposed outer surfaces, where convection with $h = 15$ W/(m^{2}K) and $T_\infty = 25°$C is imposed;
- symmetry boundaries, where homogeneous Neumann conditions are enforced.

This classification allows the Neumann and Robin boundary contributions to be assembled consistently into the global load vector and stiffness matrix. In the three-dimensional case, boundary integrals are evaluated over triangular surface facets, while in two dimensions they are evaluated over boundary edges.

8.6 Steady and Transient Solution Workflow

For steady-state problems, the assembled linear system is solved directly to obtain the nodal temperature field. For transient problems, the same spatial discretization is combined with the Backward Euler time integration scheme described in Section 7.

The transient solver advances the solution using a time step $\Delta t = 0.5$ s up to a final time $t_{\text{end}} = 1000$ s. Because all system matrices are time-independent, the left-hand matrix is assembled once and reused at every time step, resulting in an efficient and stable algorithm.

This unified implementation allows direct comparison between steady and transient results and facilitates extension from two-dimensional to three-dimensional analyses.

9 Verification and Consistency

The purpose of this section is to verify the numerical formulation and to assess the internal consistency of the finite element solutions across different dimensions, material configurations, and time integrations. Because no analytical solution exists for the present geometry and boundary conditions, verification is performed through systematic numerical checks rather than direct comparison to a closed-form solution.

9.1 Steady–Transient Consistency

A primary verification step is to compare steady-state solutions with the long-time limit of transient simulations. For linear heat conduction problems with time-independent boundary conditions, the transient solution is expected to converge to the steady solution as $t \to \infty$.

In this study, transient simulations are advanced using the Backward Euler scheme with a time step $\Delta t = 0.5$ s up to a final time $t_{\text{end}} = 1000$ s. For both single-material and thermal-paste configurations, the temperature field approaches a time-invariant state well before the final simulation time. The final transient temperature distributions are observed to match the corresponding steady-state solutions both qualitatively and quantitatively, indicating correct temporal discretization and assembly.

The monotonic convergence of the transient solution, together with the absence of non-physical oscillations, is consistent with the unconditional stability of the Backward Euler method.

9.2 Time-Step Adequacy

The selected time step $\Delta t = 0.5$ s represents a compromise between temporal resolution and computational cost. Because Backward Euler is first-order accurate in time, excessively large time steps may introduce numerical diffusion, while excessively small time steps increase cost without significant accuracy gains.

The evolution of the maximum temperature in the domain, shown in Figure **??**, exhibits a smooth and physically reasonable rise toward steady conditions. No abrupt jumps or instabilities are observed, indicating that the chosen time step is sufficiently small to resolve the dominant thermal time scales of the system.

9.3 Mesh Resolution and Spatial Consistency

Mesh verification is carried out by ensuring adequate spatial resolution in regions where strong temperature gradients are expected. In all cases, the mesh is locally refined near the tube-contact boundary and within the thermal paste layer, while coarser elements are used in regions of relatively uniform temperature.

The resulting temperature fields are smooth across element boundaries and exhibit no visible mesh-induced artifacts. In the two-material cases, temperature and heat flux continuity across the solid–paste interface is preserved, confirming correct handling of piecewise material properties during assembly.

The consistency of spatial temperature distributions between two-dimensional and three-dimensional simulations further supports the adequacy of the mesh resolution.

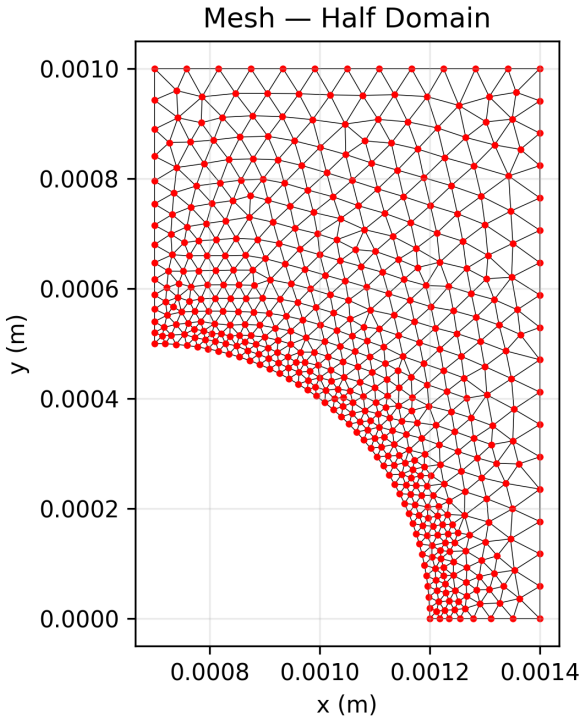

Figure 2: 2D Single Material Mesh Example Of The Half Domain.

9.4 Dimensional Extension: 2D to 3D

Another important consistency check is the extension of the formulation from two dimensions to three dimensions. The governing equations, boundary conditions, and material models remain unchanged; only the spatial discretization and boundary integration differ.

The three-dimensional steady and transient results exhibit temperature patterns that are consistent with their two-dimensional counterparts when examined on planar slices. This agreement confirms that the extrusion-based mesh construction and tetrahedral assembly preserve the physical behavior observed in the two-dimensional analysis.

9.5 Energy Balance and Boundary Consistency

The imposed heat flux $q_{\text{in}} = 1.0 \times 10^3$ W/m^2 provides a constant energy input through the tube-contact boundary. At steady state, this input is balanced by convective heat loss through the outer boundaries governed by $h = 15$ W/(m^{2}K) and $T_\infty = 25°$C.

The existence of a stable steady solution, together with convergence of transient simulations toward that solution, indicates that the global energy balance is satisfied numerically. No unbounded temperature growth is observed, further confirming the correctness of the boundary condition implementation.

9.6 Summary of Verification

The numerical results satisfy several independent verification criteria:

- convergence of transient solutions toward steady-state results;
- stability and smoothness of transient temperature evolution;
- spatial continuity across material interfaces;
- consistency between two-dimensional and three-dimensional solutions;
- physically reasonable energy balance between heat input and convective loss.

Together, these checks provide strong confidence in the correctness and robustness of the finite element implementation, allowing subsequent interpretation of the numerical results in a physically meaningful context.

10 Results and Physical Interpretation

This section presents the numerical temperature fields and explains the dominant physical mechanisms. Across all cases, heat enters through the tube-contact boundary Γ_{in} via the prescribed heat flux q_{in}, spreads by conduction inside the body (and paste when present), and is removed through convection on Γ_{conv}. To keep interpretation consistent, we emphasize (i) the hot-spot location and near-contact gradients, (ii) how the field spreads to the outer boundaries, and (iii) how transient solutions evolve toward steady behavior.

10.1 2D Steady-State Field (Single Material)

Figure 3 shows the 2D steady temperature distribution on the full (mirrored) domain. The hottest region forms on the inner arc, which is expected because the imposed Neumann boundary condition injects energy locally at Γ_{in}. From the inner arc, heat diffuses outward into the bulk and is gradually removed by convection at the outer boundary. The isotherms are densest near the tube-contact arc, indicating that the largest gradients (and therefore the largest conductive heat flux magnitudes) occur close to the heat-input boundary.

A useful interpretation is that the geometry creates a *near-contact thermal boundary layer*: the dominant temperature drop occurs over a short distance near the tube-contact region, after which the field becomes smoother in the bulk.

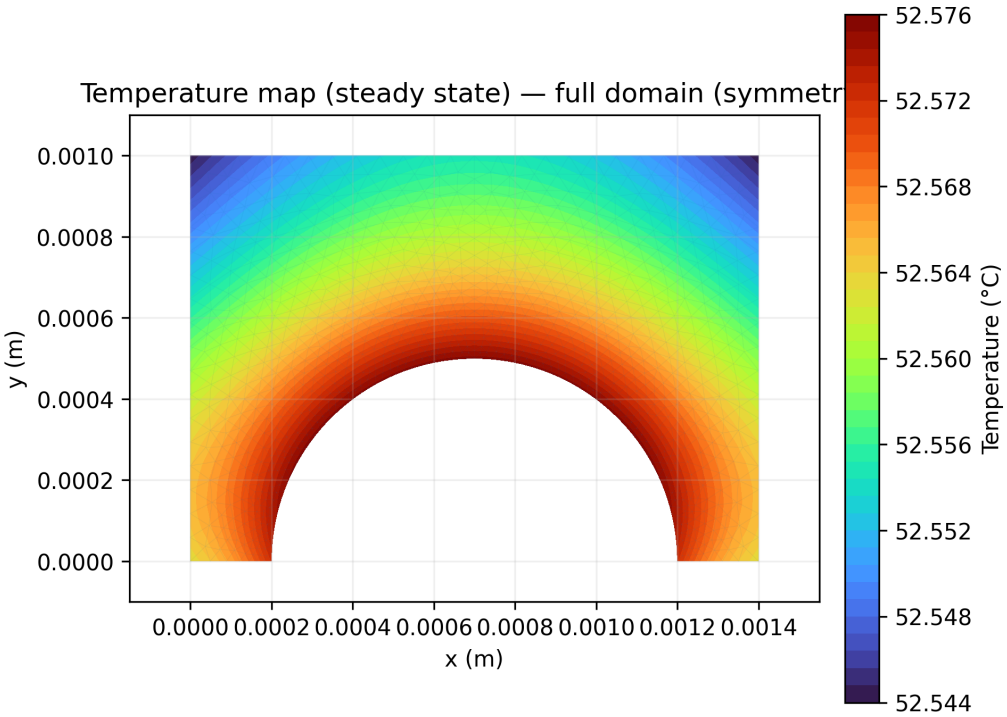

Figure 3: 2D steady-state temperature field (full mirrored domain, single material). The peak temperature occurs near the tube-contact arc, and the temperature decays toward the convective outer boundaries.

10.2 2D Transient: Early-Time Development and Approach to Steady State

To illustrate the early transient response, Figure 4 shows $\Delta T = T - T_\infty$ at $t = 0.6$ s. At this early stage, the field has only begun to develop: the plot indicates a small temperature rise with a localized increase near the tube-contact region. In the displayed frame, the temperature range is very narrow (approximately $T_{\min} \approx 25.22°\text{C}$ and $T_{\max} \approx 25.25°\text{C}$), which is consistent with diffusion starting locally before the entire domain warms significantly.

Physically, the transient evolution has two phases: (i) rapid local heating near Γ_{in} (steep gradients form first near the heat-flux boundary), followed by (ii) gradual global warming and spreading until convection removes heat at the same rate it is supplied, producing a steady solution.

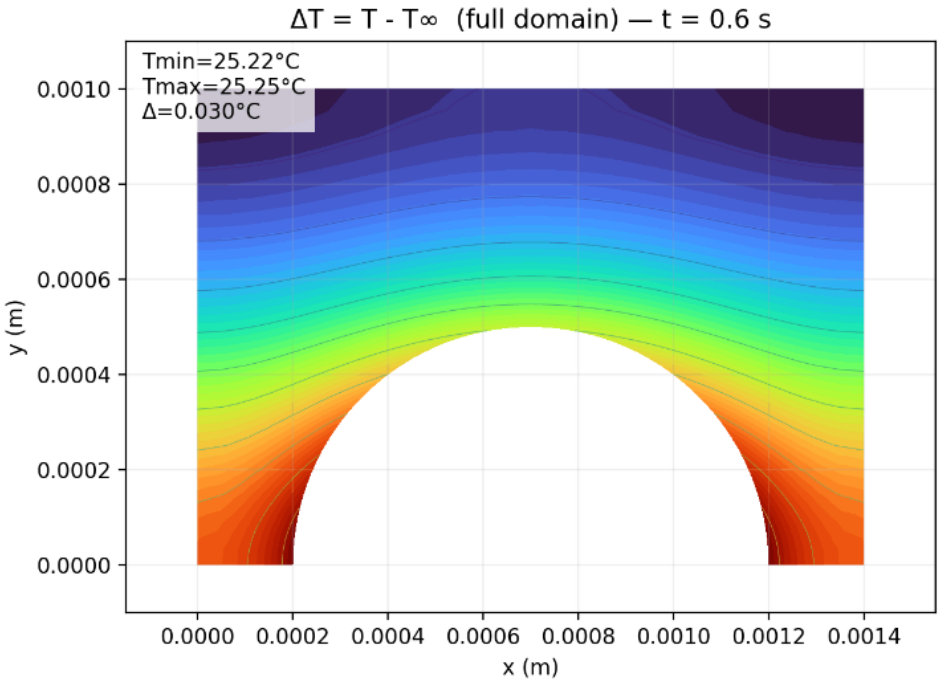

Figure 4: 2D transient field shown as $\Delta T = T - T_\infty$ at an early time ($t = 0.6$ s). The temperature rise is initially localized near the tube-contact arc and remains small throughout the domain at very early times.

10.3 2D Thermal Paste Case: Mesh Resolution and Near-Contact Gradients

The thermal paste layer is a thin annulus around the tube-contact arc, and its thickness is small compared to the body dimensions.From a heat-transfer perspective, the thermal paste layer acts as a localized thermal resistance between the tube-contact boundary and the bulk solid. This behavior is consistent with classical descriptions of thermal contact resistance, where thin interfacial layers can significantly influence local temperature gradients even when their thickness is small relative to the overall geometry [6].

Therefore, it must be meshed with local refinement to avoid artificially smearing the temperature gradient across the layer. Figure 5 shows the full-domain (mirrored) mesh for the two-material case, where refinement is concentrated around the paste region.

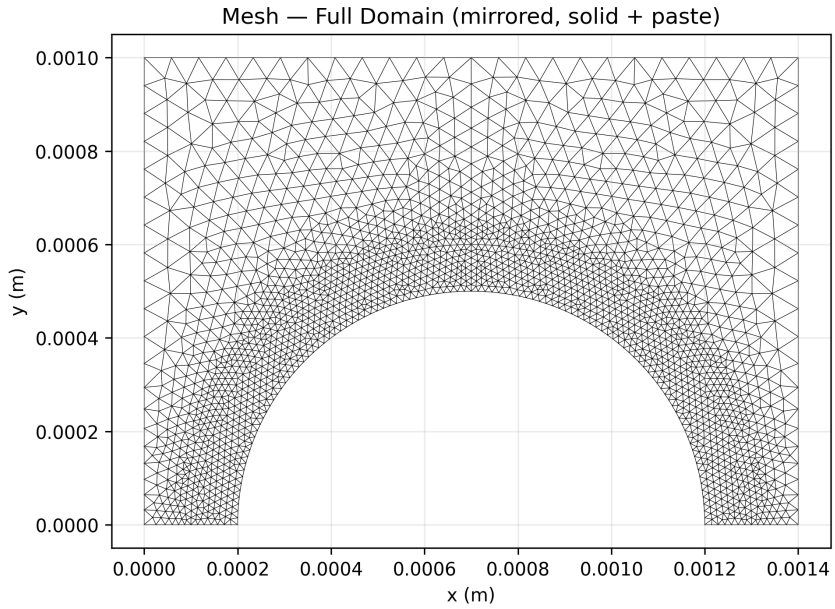

Figure 5: 2D mesh for the two-material (solid + paste) case on the full mirrored domain. Strong local refinement is used around the thin paste annulus to resolve near-contact gradients.

Figure 6 shows a transient snapshot for the paste case ($t = 11.60$ s) plotted as ΔT. The near-contact region develops a sharper thermal boundary layer because the paste modifies the effective conduction path between the tube-contact boundary and the bulk. In a resistance sense, the paste contributes an additional local thermal resistance, so for the same heat flux the temperature drop concentrates more strongly near the interface region, which is visible as tighter contours near the inner arc.

(The frame shows a narrow temperature spread across the domain at that instant, with values around $T \approx 28.1°\mathrm{C}$, i.e., a few degrees above ambient, while the spatial variation within the snapshot remains small compared to the mean rise.)

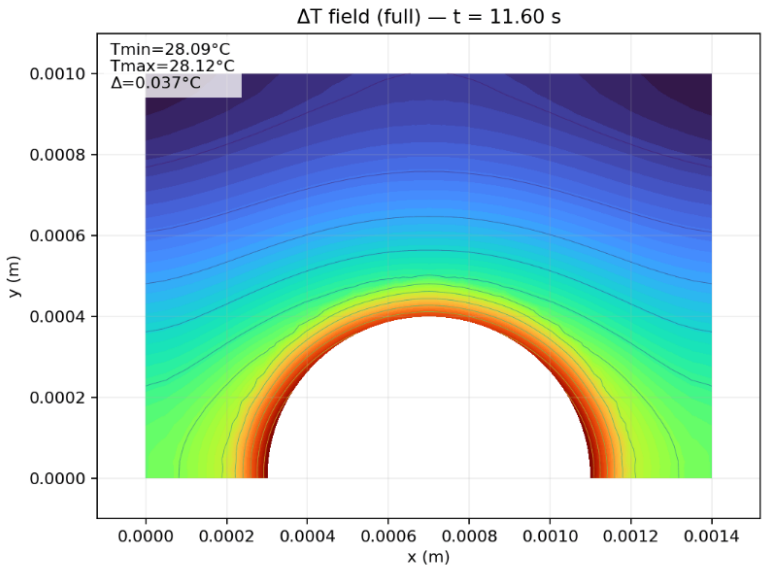

Figure 6: 2D transient snapshot for the thermal paste case ($t = 11.60$ s) shown as $\Delta T = T - T_\infty$. The paste layer concentrates the steepest gradients near the tube-contact region, consistent with an added local thermal resistance.

10.4 3D Geometry and Mesh

The 3D model is constructed by extruding the 2D cross-section and discretizing the volume with tetrahedral elements. Figure 7 shows the full (mirrored) 3D mesh. The mesh density inherited from the refined 2D cross-section ensures adequate resolution near the tube-contact surface, while the volume discretization enables internal slicing for interpretation.

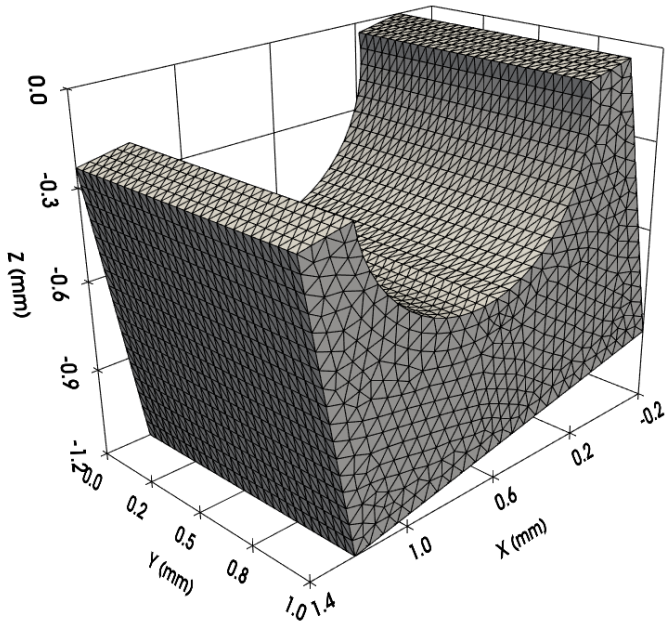

Figure 7: Full (mirrored) 3D tetrahedral mesh used for the steady and transient simulations. The tube-contact cut is resolved with finer surface triangulation to capture near-contact gradients.

10.5 3D Steady Field: Surface View and Interior Slices

Figure 8 shows the 3D steady-state temperature mapped on the outer surface. As in 2D, the hottest region is located at the tube-contact boundary, while temperatures decrease toward the outer convective surfaces. The 3D visualization makes the role of the exposed area more apparent: convection acts over multiple faces, which promotes cooling away from the tube-contact region.

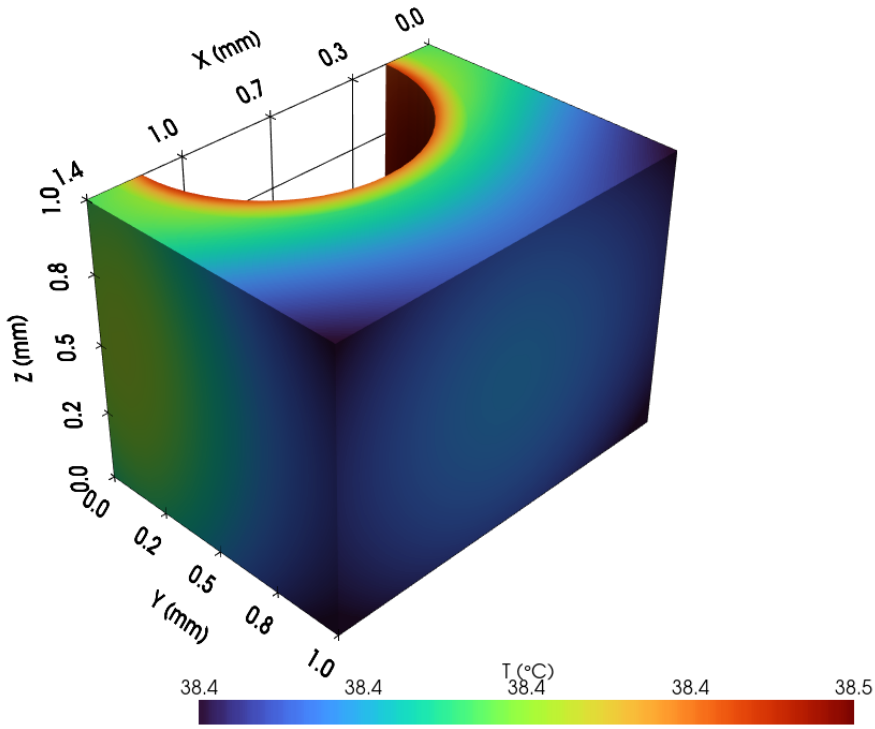

Figure 8: 3D steady-state temperature distribution (surface rendering). The hot spot remains localized near the tube-contact region, while outer faces approach lower temperatures due to convection.

To inspect internal gradients (which surface plots can hide), Figure 9 presents an internal slice. The slice confirms that the strongest gradients occur adjacent to the tube-contact surface and decay into the bulk. In other words, the principal thermal bottleneck is near the contact region; once heat spreads into the bulk, convection removes it over the outer surfaces.

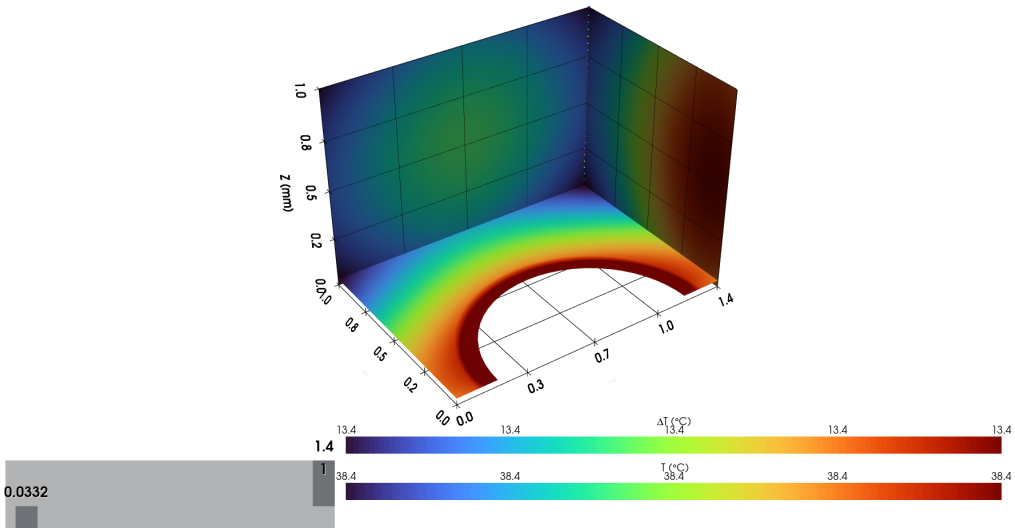

Figure 9: Internal slice of the 3D steady-state temperature field. Slice visualization highlights the steep near-contact gradients and the smoother bulk temperature variation.

10.6 3D Transient Field at Final Time

Figure 10 shows the 3D transient temperature field at the final simulated time. The spatial pattern matches the steady case: the same hot-spot location and decay toward the outer boundaries are observed. This agreement is the expected behavior for time-independent boundary conditions, because the transient solution converges to the steady solution as $t \to \infty$.

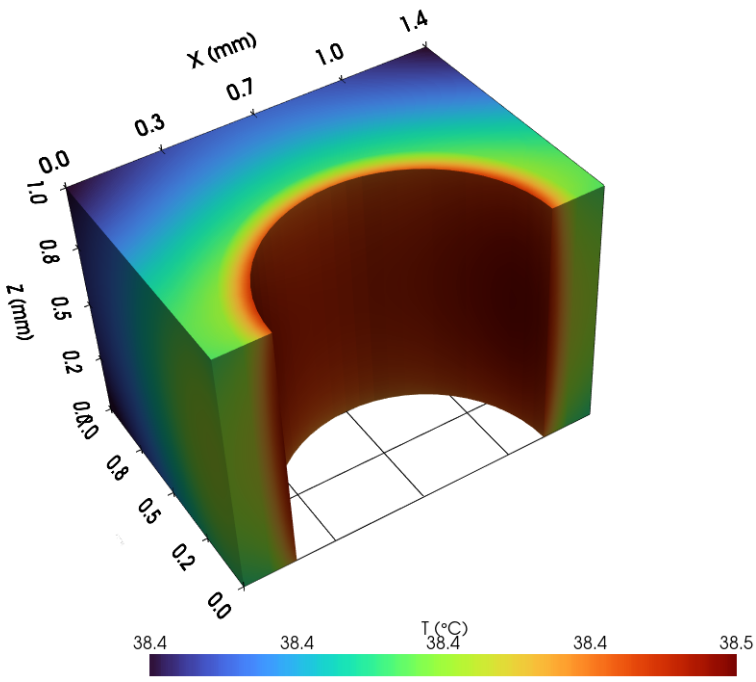

Figure 10: 3D transient temperature field at the final simulated time (surface rendering). The field exhibits the same qualitative structure as the steady solution, indicating convergence.

An internal transient slice is shown in Figure 11. It confirms smooth internal gradients and the persistence of the near-contact thermal boundary layer in time-dependent evolution.

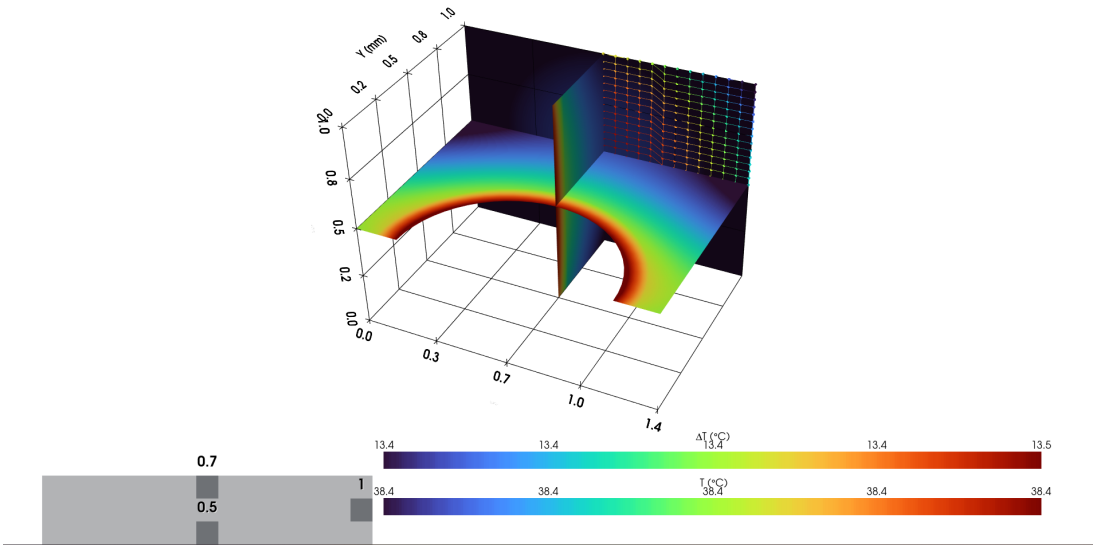

Figure 11: Internal slice of the 3D transient temperature field at the final simulated time. The slice confirms near-contact gradients and overall convergence toward a steady pattern under constant boundary conditions.

10.7 Key Physical Takeaways (All Cases)

The simulations support the following physical interpretation:

- **Hot spot location:** The maximum temperature occurs at or immediately adjacent to the tube-contact boundary, because heat is injected there through the prescribed heat flux.
- **Near-contact thermal boundary layer:** The steepest gradients concentrate close to Γ_{in}, indicating that most of the temperature drop occurs near the contact region rather than uniformly across the domain.
- **Effect of thermal paste:** The paste layer modifies the effective conduction path and concentrates gradients near the tube-contact interface, consistent with an added local thermal resistance.
- **Transient-to-steady consistency:** Under time-independent forcing, transient solutions evolve smoothly toward a steady temperature field with matching qualitative structure (hot spot near contact, decay toward convective boundaries).
- **3D insight:** 3D rendering and internal slices provide information not accessible in experiments at this scale, especially the internal gradient structure near the tube-contact region, which is the most relevant area for interpreting thermocouple attachment behavior.

11 Discussion

Due to the curved geometry, mixed boundary conditions, and presence of a thin multi-material interface, analytical solutions are not available, making FEM a necessary and appropriate tool for this problem. The numerical results demonstrate how heat introduced through the tube-contact boundary is redistributed by conduction and ultimately removed by convection at the exposed surfaces. The imposed heat flux $q_{\text{in}} = 1.0 \times 10^3$ W/m^2 generates a localized hot spot near the tube-contact region in all cases, while the Robin boundary condition with $h = 15$ W/(m^{2}K) and $T_\infty = 25°$C provides a stabilizing heat removal mechanism that prevents unbounded temperature growth.

A key observation is the systematic difference between two-dimensional and three-dimensional results. The 2D formulation represents an idealized per-unit-depth extrusion and therefore neglects convective heat loss from out-of-plane surfaces. As a consequence, the effective convective area in 2D is smaller, leading to higher steady-state temperatures. In contrast, the 3D simulations introduce additional exposed surfaces in the axial direction and allow heat to spread three-dimensionally, increasing the total convective heat loss. The lower temperature levels observed in the 3D cases are therefore a direct and physically expected consequence of increased heat dissipation rather than a numerical artifact.

Transient simulations performed using the implicit Backward Euler scheme ($\Delta t = 0.5$ s, $t_{\text{end}} = 1000$ s) exhibit smooth, monotonic convergence toward steady-state solutions. This behavior is consistent with diffusion-dominated heat transfer and confirms the numerical stability of the time integration scheme. Transient results also provide insight into the thermal time scales of the system, showing rapid local heating near the tube-contact boundary followed by slower global equilibration as convective losses become dominant.

The inclusion of a thermal paste layer ($t_p = 0.1$ mm) modifies the effective conduction path between the tube-contact surface and the bulk solid. The paste introduces a localized thermal resistance that concentrates temperature gradients near the contact region and increases local temperatures for a given heat flux.

This effect is physically consistent with classical thermal resistance arguments and highlights the importance of accurately resolving thin interfacial layers in micro-scale heat-transfer models.

Overall, the simulations clarify how geometry, dimensionality, and interface properties collectively control temperature distributions in microtube-contact configurations. The ability to visualize internal temperature fields—particularly in three dimensions—provides valuable insight that is difficult to obtain experimentally at this scale.

Table 1: Comparison of characteristic temperatures for different modeling cases.

Case	$T_{\max}$ (°C)	$T_{\min}$ (°C)	$\Delta T_{\max}$ (°C)
2D steady (single material)	~ 52.6	~ 52.5	~ 27.6
2D steady (with paste)	higher than single	—	increased
3D steady (with paste)	~ 38.5	~ 38.4	~ 13.5

The observed differences between two-dimensional and three-dimensional temperature levels are consistent with established dimensional considerations in heat transfer modeling. Lower-dimensional models restrict available heat dissipation paths and therefore tend to overpredict temperatures when compared to fully three-dimensional analyses that include additional convective surfaces [2, 7].

12 Conclusions

A finite element framework was developed to study heat conduction in a microtube-contact geometry under steady and transient conditions. The model incorporates prescribed heat flux (Neumann), convective heat loss (Robin), and symmetry boundary conditions, and it is extended naturally to multi-material domains through piecewise material properties.

Two-dimensional simulations provide an efficient baseline description of near-contact heat transfer and highlight the formation of strong local temperature gradients at the tube-contact boundary. However, comparison with three-dimensional results demonstrates that 2D models inherently underestimate convective heat loss because they neglect out-of-plane surfaces. Three-dimensional simulations introduce additional heat dissipation pathways and axial heat spreading, resulting in lower and more physically representative temperature levels.

Transient analyses using an implicit Backward Euler scheme show stable and monotonic convergence toward steady-state solutions and allow direct examination of thermal time scales. The inclusion of a thin thermal paste layer significantly alters near-contact temperature gradients by introducing a localized thermal resistance, emphasizing the importance of interface modeling in micro-scale thermal systems.

The combined 2D and 3D results illustrate how dimensionality, boundary conditions, and interface properties interact to control temperature fields in microtube-contact geometries. The numerical framework and visualization techniques developed here provide a useful tool for interpreting experimental measurements and for guiding the design and analysis of micro-scale thermal contact configurations.

References

[1] Hojin Ahn, G. Oğuzhan, and Ö. Efe. Experimental thermal characterization of microtube heat transfer using molded solder contact techniques. *Applied Thermal Engineering*, 211:118435, 2022. doi: 10.1016/j.applthermaleng.2022.118435. URL `https://www.sciencedirect.com/science/article/pii/S1359431122002587`.

[2] F. P. Incropera, D. P. DeWitt, T. L. Bergman, and A. S. Lavine. *Fundamentals of Heat and Mass Transfer*. John Wiley & Sons, 7 edition, 2011.

[3] M. N. Özişik. Transient heat conduction in finite bodies. *International Journal of Heat and Mass Transfer*, 23(1):1–11, 1980. doi: 10.1016/0017-9310(80)90002-3.

[4] O. C. Zienkiewicz. The weighted residual finite element method in engineering. *International Journal for Numerical Methods in Engineering*, 11(1):1–24, 1977. doi: 10.1002/nme.1620110102.

[5] R. W. Lewis and K. Morgan. Finite element methods in heat transfer analysis. *Applied Mathematical Modelling*, 20(6):430–442, 1996. doi: 10.1016/0307-904X(95)00137-X.

[6] M. M. Yovanovich. Thermal contact resistance across real interfaces. *Journal of Heat Transfer*, 128(4):305–317, 2005.

[7] T. L. Bergman. Dimensional considerations in heat transfer modeling. *Journal of Heat Transfer*, 133(3):031005, 2011. doi: 10.1115/1.4002475.